Advanced Network Project

Technical Documentation

Prepared by:

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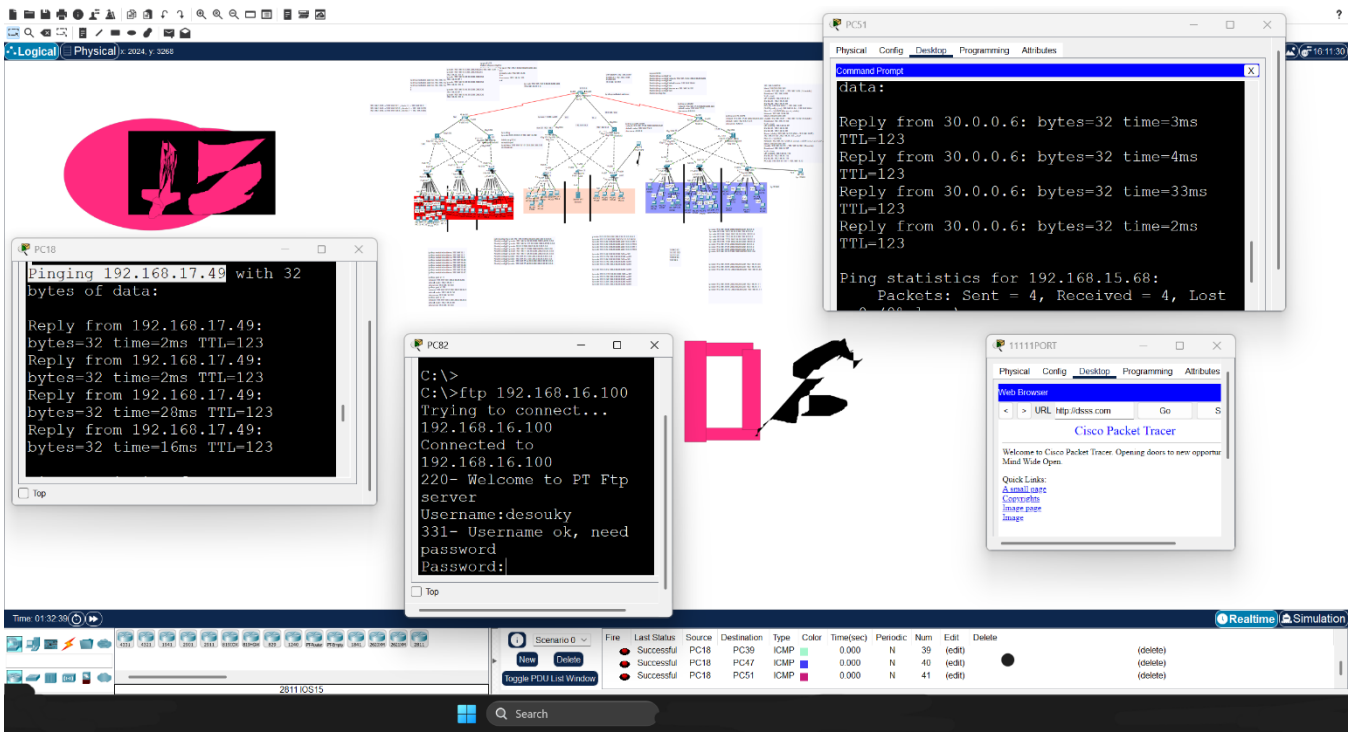
Network Scope

3 Buildings | 3 Floors Each | 9 Subnets

6 Multilayer Switches | 9 Access Switches | 3 Edge Routers

1. Project Overview

This project involves designing and implementing a complete enterprise network spanning 3 buildings, each with 3 floors. The network infrastructure provides full connectivity, redundancy, and security across all locations, serving all devices with automated IP assignment and access to centralized services.



1.1 Network Infrastructure Summary

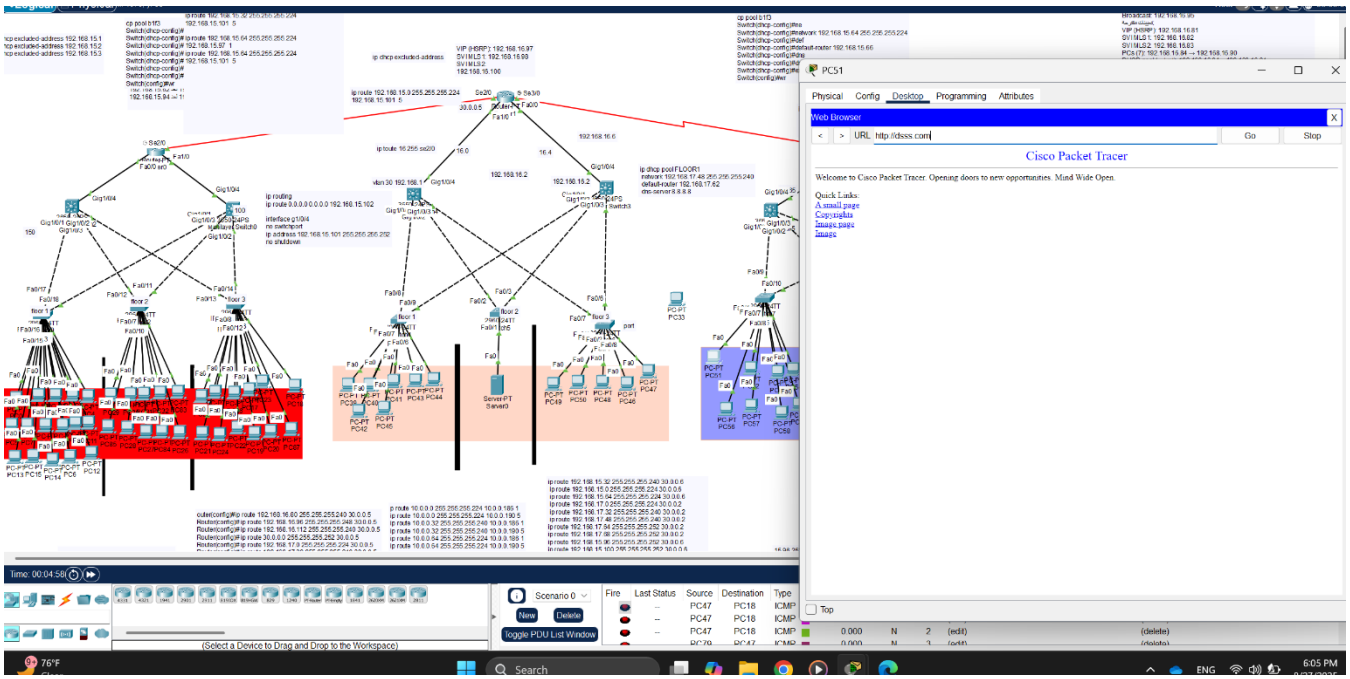
Component	Quantity	Role
Multilayer Switches	6	Inter-VLAN routing & aggregation per building
Access Switches	9	End-device connectivity (1 per floor)
Edge Routers	3	WAN/Internet connectivity & NAT (1 per building)
Servers (per building)	Multiple	DHCP, DNS, FTP, HTTP, File Sharing
VLANs	9	One per floor for traffic segmentation

1.2 Design Goals

- High availability through redundant links and gateway protocols (HSRP/VRRP)
- Security through VLAN segmentation, ACLs, and port security

- Scalability with structured IP addressing per building and floor
- Full service coverage: DHCP, DNS, FTP, HTTP, and file sharing in every building

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2. IP Addressing & DHCP Configuration

DHCP servers are distributed across all three buildings to automatically assign IP addresses. Each floor has a dedicated DHCP pool with its own subnet to prevent address conflicts and organise traffic efficiently.

2.1 Subnet Plan

Building	Floor	Network	Subnet Mask	Gateway	DNS Server
Building 1	Floor 1	192.168.15.0	255.255.255.240	192.168.15.1	192.168.15.10
Building 1	Floor 2	192.168.16.0	255.255.255.240	192.168.16.1	192.168.16.10
Building 1	Floor 3	192.168.17.0	255.255.255.240	192.168.17.1	192.168.17.10
Building 2	Floor 1	192.168.18.0	255.255.255.240	192.168.18.1	192.168.18.10
Building 2	Floor 2	192.168.19.0	255.255.255.240	192.168.19.1	192.168.19.10
Building 2	Floor 3	192.168.20.0	255.255.255.240	192.168.20.1	192.168.20.10
Building 3	Floor 1	192.168.21.0	255.255.255.240	192.168.21.1	192.168.21.10
Building 3	Floor 2	192.168.22.0	255.255.255.240	192.168.22.1	192.168.22.10
Building 3	Floor 3	192.168.23.0	255.255.255.240	192.168.23.1	192.168.23.10

2.2 DHCP Pool Configuration Example

The following is a sample DHCP pool configuration applied on the Building 1, Floor 1 server:

```
ip dhcp pool Building1_Floor1
  network 192.168.15.0 255.255.255.240
  default-router 192.168.15.1
  dns-server 192.168.15.10

ip dhcp pool Building1_Floor2
  network 192.168.16.0 255.255.255.240
  default-router 192.168.16.1
  dns-server 192.168.16.10

ip dhcp pool Building1_Floor3
  network 192.168.17.0 255.255.255.240
  default-router 192.168.17.1
  dns-server 192.168.17.10
```

3. VLAN Configuration & Inter-VLAN Routing

VLANs are created for each floor across all buildings to segment network traffic and improve both performance and security. Inter-VLAN routing is performed by the Multilayer Switches, eliminating the need for a dedicated router for internal routing.

3.1 VLAN Assignment

VLAN ID	Name	Assigned To	Network
VLAN 10	Building1_Floor1	Building 1 – Floor 1	192.168.15.0/28
VLAN 20	Building1_Floor2	Building 1 – Floor 2	192.168.16.0/28
VLAN 30	Building1_Floor3	Building 1 – Floor 3	192.168.17.0/28
VLAN 40	Building2_Floor1	Building 2 – Floor 1	192.168.18.0/28
VLAN 50	Building2_Floor2	Building 2 – Floor 2	192.168.19.0/28
VLAN 60	Building2_Floor3	Building 2 – Floor 3	192.168.20.0/28
VLAN 70	Building3_Floor1	Building 3 – Floor 1	192.168.21.0/28
VLAN 80	Building3_Floor2	Building 3 – Floor 2	192.168.22.0/28
VLAN 90	Building3_Floor3	Building 3 – Floor 3	192.168.23.0/28

3.2 Static Routing Configuration

Static routes are configured on the Multilayer Switches to ensure full connectivity between all floors, buildings, and servers:

```
! Building 1 - inter-floor static routes
ip route 192.168.15.0 255.255.255.240 192.168.15.1
ip route 192.168.16.0 255.255.255.240 192.168.16.1
ip route 192.168.17.0 255.255.255.240 192.168.17.1

! Building 2 - inter-floor static routes
ip route 192.168.18.0 255.255.255.240 192.168.18.1
ip route 192.168.19.0 255.255.255.240 192.168.19.1
ip route 192.168.20.0 255.255.255.240 192.168.20.1

! Building 3 - inter-floor static routes
ip route 192.168.21.0 255.255.255.240 192.168.21.1
ip route 192.168.22.0 255.255.255.240 192.168.22.1
ip route 192.168.23.0 255.255.255.240 192.168.23.1
```

4. Access Control Lists (ACL)

Extended ACLs are applied to control access to specific services such as FTP, restricting or permitting traffic on a per-host basis. The ACLs are applied inbound on the relevant interface.

4.1 FTP Access Control Example

The following ACL allows one specific host to access the FTP server while denying access to another:

```
! ACL 101 - FTP access control
access-list 101 permit tcp host 192.168.15.2 host 192.168.15.100 eq ftp
access-list 101 deny   tcp host 192.168.15.3 host 192.168.15.100 eq ftp
access-list 101 permit ip any any

! Apply the ACL inbound on interface GigabitEthernet 0/0
interface GigabitEthernet 0/0
  ip access-group 101 in
```

4.2 ACL Rule Summary

Rule	Source	Destination	Service	Action
1	192.168.15.2	192.168.15.100	FTP (TCP 21)	Permit
2	192.168.15.3	192.168.15.100	FTP (TCP 21)	Deny
3	Any	Any	All	Permit

5. Port Security

Port security is enabled on Access Switch interfaces to restrict which devices can connect to the network. Using the sticky MAC address feature, the switch automatically learns and locks the MAC addresses of connected devices.

5.1 Port Security Configuration

```
interface FastEthernet 0/1
  switchport mode access
  switchport port-security
  switchport port-security maximum 2
  switchport port-security violation restrict
  switchport port-security mac-address sticky
```

5.2 Parameter Explanation

Parameter	Value	Description
maximum	2	Maximum number of MAC addresses allowed on the port
violation	restrict	Drops violating frames and logs a security warning; port stays up
mac-address sticky	Enabled	Dynamically learns & saves MAC addresses to running config

6. NAT Configuration

Network Address Translation (NAT) is configured on all three Edge Routers to allow internal devices to access external networks and the Internet using a shared public IP address.

6.1 NAT Configuration Example

```
! Define inside and outside interfaces
interface GigabitEthernet 0/0
  ip nat inside
  ip address 192.168.15.1 255.255.255.240

interface GigabitEthernet 0/1
  ip nat outside
  ip address <public-ip> 255.255.255.0

! Define the ACL for NAT (which internal traffic to translate)
access-list 1 permit 192.168.15.0 0.0.0.15
access-list 1 permit 192.168.16.0 0.0.0.15
access-list 1 permit 192.168.17.0 0.0.0.15

! Enable PAT (NAT overload)
ip nat inside source list 1 interface GigabitEthernet 0/1 overload
```

7. Servers & Network Services

Servers are deployed in all three buildings to guarantee service availability and eliminate single points of failure. Each building hosts the following services:

Service	Protocol/Port	Purpose	Availability
DHCP	UDP 67/68	Automatic IP address assignment to all end devices	All Buildings
DNS	UDP/TCP 53	Domain name to IP address resolution	All Buildings
FTP	TCP 21	Secure file transfer between devices and servers	All Buildings
HTTP	TCP 80	Internal and external web services hosting	All Buildings
File Sharing	SMB/CIFS	Shared network drives and resource access	All Buildings

8. Redundancy & High Availability

The network is designed with redundancy at every layer to ensure continuous availability even during device failures or link outages.

8.1 Physical Redundancy

- Redundant uplinks between Access Switches and Multilayer Switches
- Dual connections from Multilayer Switches to Edge Routers
- Spanning Tree Protocol (STP) prevents switching loops while maintaining redundant paths

8.2 Gateway Redundancy – HSRP/VRRP

Hot Standby Router Protocol (HSRP) or Virtual Router Redundancy Protocol (VRRP) is configured on the Multilayer Switches to provide a virtual gateway IP for each VLAN. If the active gateway fails, the standby takes over within seconds.

```
! HSRP configuration on Multilayer Switch - VLAN 10
interface Vlan10
  ip address 192.168.15.2 255.255.255.240
  standby 10 ip 192.168.15.1
  standby 10 priority 110
  standby 10 preempt

! Standby Multilayer Switch - VLAN 10
interface Vlan10
  ip address 192.168.15.3 255.255.255.240
  standby 10 ip 192.168.15.1
  standby 10 priority 100
  standby 10 preempt
```

8.3 Redundancy Summary

Mechanism	Layer	Benefit
Redundant Links	Physical / Data Link	No single link failure can isolate a floor or building
STP	Data Link	Loop prevention with automatic failover path activation
HSRP/VRRP	Network	Virtual gateway ensures end devices always reach the router
Distributed Servers	Application	No single server outage disrupts all services

9. Network Design Summary

The network designed by Mohamed Ibrahim Eldesouky delivers a robust, secure, and highly available enterprise infrastructure. Key design highlights are summarised below:

Area	Implementation
Infrastructure	6 Multilayer Switches, 9 Access Switches, 3 Edge Routers across 3 buildings
IP Addressing	Structured /28 subnets per floor (192.168.15.0 – 192.168.23.0)
DHCP	Distributed DHCP servers with per-floor pools in all buildings
VLAN Segmentation	9 VLANs (VLAN 10–90), one per floor; inter-VLAN routing on Multilayer Switches
Routing	Static routes on Multilayer Switches for full inter-floor/inter-building connectivity
ACL	Extended ACLs controlling service access (e.g. FTP) per host
Port Security	Sticky MAC, max 2 addresses, restrict violation mode on Access Switch ports
NAT	PAT (overload) on all Edge Routers for Internet access
Services	DHCP, DNS, FTP, HTTP, and file sharing distributed across all buildings
Redundancy	Redundant links + STP + HSRP/VRRP for gateway failover

The result is a network engineered for availability, security, and efficient traffic flow across all buildings and floors, meeting all enterprise requirements.